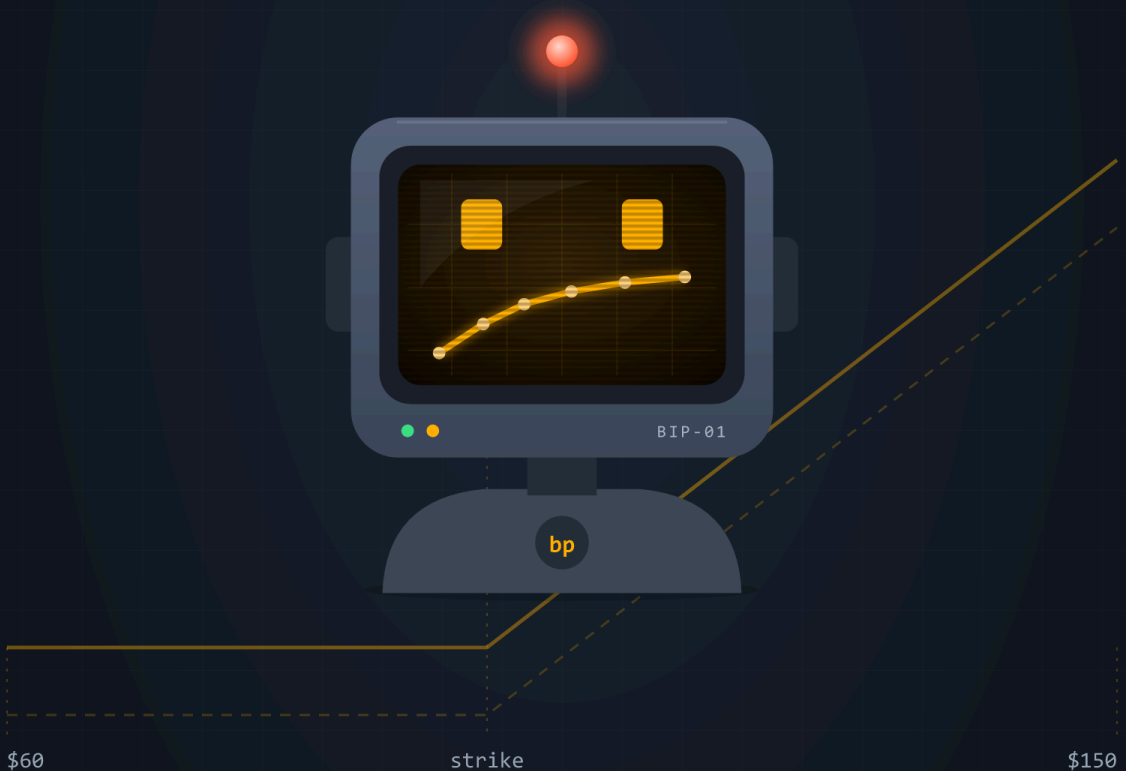


OPTIONS FROM ZERO • PART 1

Options from Zero Part 1

From payoffs to the Black-Scholes
formula, with Python and QuantLib



Rates with Bip

OPTIONS FROM ZERO • PART 1 OF 3

Options from Zero

Part 1: From payoffs to the Black-Scholes formula, with Python and QuantLib

Rates with Bip

Options from Zero. Part 1: From payoffs to the Black-Scholes formula, with Python and QuantLib.

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The share in this book, its option prices, its volatility and the interest rate are illustrative, made up for teaching. They are not market data. The AUD/USD exchange rates in Chapter 9 are the noon buying rates in New York from the Federal Reserve Board's H.10 release. Source: Board of Governors of the Federal Reserve System.

Facts about listed options are checked against ASX's published material, and the history in Chapter 10 against the Royal Swedish Academy of Sciences' 1997 press release; sources are listed at the end of each chapter. Options are risky: a seller can lose far more than the premium received.

Every number in the text, tables and figures is computed by the companion QuantLib Python notebooks (QuantLib 1.43), one per chapter, which reproduce each result.

This book is for education only. It is not financial, investment, legal or tax advice.

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How to use this book

This book follows the ten video episodes of Part 1 of *Options from Zero*, one chapter per episode, and goes a little deeper in each. It assumes nothing: no options, no calculus, and no probability beyond "a coin has two sides". By the end you will be able to derive the price of an option on a tree by hand, read every term of the Black-Scholes formula, and check both against QuantLib.

One question runs through the whole of Part 1. A share costs \$100 today. What is the right, but not the obligation, to buy it for \$105 in a year worth? Chapter 1 gives the answer, \$7.39, and admits we can't explain it yet. Chapter 10 derives exactly that number. Everything in between is the road from one to the other.

- **Chapter 1** says what options are: calls, puts, payoff and profit.
- **Chapter 2** is a maths pit stop on e , discounting and logarithms.
- **Chapters 3–6** are the heart of the book: why an option's price is not its expected payoff, put-call parity, copying an option with shares and a loan, and risk-neutral probability.
- **Chapter 7** builds real binomial trees and prices American options.
- **Chapter 8** is a maths pit stop on the bell curve.
- **Chapter 9** says what volatility is, with real AUD/USD data.
- **Chapter 10** decodes the Black-Scholes formula.

Each chapter ends with the chapter's one idea to remember, which the videos call *Bip's rule*, and a question to try before the next chapter. The answer opens the next chapter. The appendix has a glossary and the Part 1 checkpoint with worked answers.

Numbers you can reproduce

Every number in the text, tables and figures is computed by a companion Python notebook, one per chapter; nothing is typed in by hand. Where a chapter works an example by hand, it also shows that QuantLib gets the same answer. You are encouraged to run the notebooks, change the inputs and watch the results move.

The notebooks are free to download from github.com/RatesWithBip/curriculum, in the `options-from-zero` folder, and each has an *Open in Colab* button. They run as they are in Google Colab: each installs QuantLib and writes its own data, so there is nothing else to download.

About the data

The share in this book is made up: it costs \$100, pays no dividends and has a volatility of 20%, and the interest rate is 3.60% a year. Real option prices are licensed data, and the ideas don't depend on the exact levels. The one exception is Chapter 9, which measures the volatility of the Australian dollar from the Federal Reserve Board's published exchange rates.

What comes next

Part 2 derives the same formula a second way, by hedging, and turns to the Greeks, Monte Carlo, implied volatility, and options on AUD/USD and on BBSW. Part 3 takes apart the assumption that volatility is constant.

Free, and staying free

The videos, the notebooks and this book are free. If they help you, you can support the channel with a coffee at ko-fi.com/rateswithbip.

Options are risky. This book is for education only and is not financial advice.

Options Are Insurance

Calls and puts, strike, expiry and premium. What an option pays, what it earns you after its price, and who carries the risk.

Rights, not obligations

Think of home insurance. You pay a premium, and you only claim if something goes wrong. Options work the same way.

- A **call** gives its buyer the right, but not the obligation, to buy a share at a fixed price, the **strike**, up to a date called **expiry**.
- A **put** gives the right to sell at the strike.
- Either way, the buyer pays a **premium** up front for the right.

That is how ASX describes the options listed on its market. The right is the whole point: you use it only when it pays, and otherwise let it lapse.

The running trade

Throughout Part 1 we use one example. A share costs \$100 today and pays no dividends. The option has a strike of \$105 and expires in one year. Two more inputs will matter later: the share's volatility, 20%, and the interest rate, 3.6% a year.

The call costs \$7.39 and the put \$8.68. Those two numbers come from the Black-Scholes formula. We take them as given for now: explaining them is the job of the next nine chapters.

Payoff

At expiry the call is worth

$$C_T = \max(S_T - K, 0)$$

where S_T is the share price at expiry and K the strike. Below the strike you would never use the right to buy at \$105, so the call pays nothing. Above it, the call pays a dollar for every dollar the share is over the strike. The put is the mirror image, $\max(K - S_T, 0)$: it pays when the share falls.

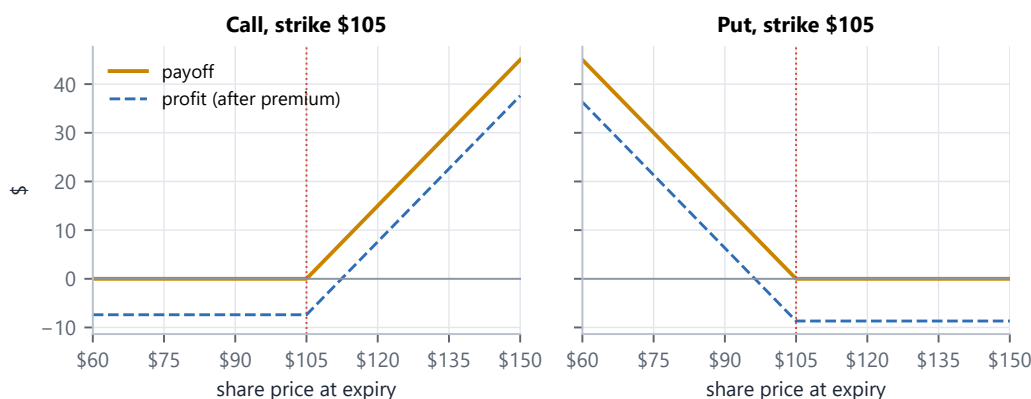


Figure 1.1. Payoff and profit at expiry for the one-year call and put, strike \$105. The dotted line marks the strike. Illustrative share.

Payoff is not profit

The buyer paid the premium at the start, so the profit is the payoff minus the premium. On a payoff diagram the whole line drops by the premium. (Strictly, the premium was paid a year earlier and could have earned interest; payoff diagrams conventionally ignore that, and so do we here.)

Payoff and profit at expiry, strike \$105. Call premium \$7.39, put premium \$8.68.

SHARE AT EXPIRY	CALL PAYOFF	CALL PROFIT	PUT PAYOFF	PUT PROFIT
\$80	\$0.00	-\$7.39	\$25.00	+\$16.32
\$90	\$0.00	-\$7.39	\$15.00	+\$6.32
\$100	\$0.00	-\$7.39	\$5.00	-\$3.68
\$105	\$0.00	-\$7.39	\$0.00	-\$8.68
\$110	\$5.00	-\$2.39	\$0.00	-\$8.68
\$120	\$15.00	+\$7.61	\$0.00	-\$8.68
\$130	\$25.00	+\$17.61	\$0.00	-\$8.68

The call's **break-even** is the strike plus the premium, \$112.39; the put's is the strike minus its premium, \$96.32. Below the call's break-even the buyer loses money, and below the strike the whole premium is gone. That happens whenever an option expires

worthless, so a bought option is not "safer" than the share: it can lose everything paid for it.

Why a put is insurance

If you own the share and buy the put, you can always sell at \$105 however far the share falls. Share plus put is worth at least the strike at expiry, and the premium is the cost of that cover. At share prices of \$50, \$80, \$105 and \$130, share plus put is worth \$105, \$105, \$105 and \$130.

Notice that the put costs more than the call. That isn't fear. Chapter 4 shows it comes entirely from where the strike sits relative to the forward price.

Two sides

Every option has a buyer, which ASX calls the **taker**, and a seller, the **writer**. The writer collects the premium and takes on the obligation. Whatever one makes, the other loses: their profits add up to zero. The taker can lose at most the premium. The writer of a call can lose far more, with no limit in principle, because a share price has no ceiling. If you sold our call for \$7.39 and the share ended at \$150, you would lose \$37.61.

A straddle

Options combine. Buy a call and a put with the same strike, \$100, for \$15.90 in total. At expiry the pair pays the distance between the share and the strike, in either direction. If the share ends at \$80 or \$120 you make \$4.10 either way; if it barely moves you lose up to the full cost. The straddle breaks even at \$84.10 and \$115.90.

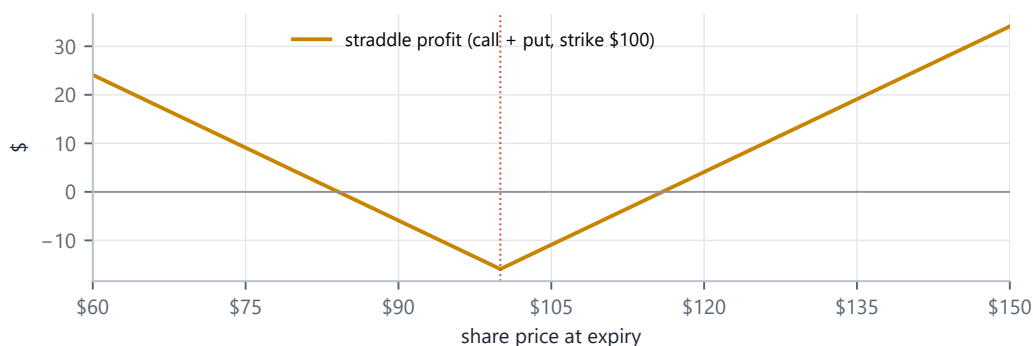


Figure 1.2. Profit at expiry of a straddle: a call and a put, both struck at \$100, after both premiums.

A straddle doesn't care which way the share goes, only how far. That is our first meeting with **volatility**, the idea that runs through the whole series.

In the notebook

The notebook computes the premiums with the formula in `options.py` and checks them against QuantLib's analytic Black-Scholes engine: the call is \$7.39 and the put \$8.68 either way.

The same two premiums from QuantLib

```
for name, kind in (("call", ql.Option.Call), ("put", ql.Option.Put)):
    opt = ql.VanillaOption(ql.PlainVanillaPayoff(kind, K), ql.EuropeanExercise(expiry))
    opt.setPricingEngine(ql.AnalyticEuropeanEngine(process))
    ql_price[name] = opt.NPV()
out["quantlib"] = {"call": ql_price["call"], "put": ql_price["put"], "years":
    dc.yearFraction(today, expiry),
```

Bip's rule. Payoff is what you get. Profit is what you keep.

Try this. You sold the call for \$7.39. The share ends at \$130. What is your profit?

Sources

Companion notebook: `s3e01_options_are_insurance.ipynb`. Every number in this chapter is computed by it.

1. ASX, *Options course, Module 2: What are options*. <https://www.asx.com.au/investors/investment-tools-and-resources/online-courses/options-course-quiz/what-are-options>
2. ASX, *Understanding Options Trading*. <https://www.asx.com.au/content/dam/asx/investors/investment-options/options/understanding-options.pdf>

Maths Pit Stop: exp and log

Where the number e comes from, why it grows money and discounts it, and how logarithms turn growth into returns that add up.

Last chapter's question

You sold the call for \$7.39 and the share ended \$130. You owe the buyer the payoff, \$25.00, so your profit is −\$17.61. Selling options can hurt.

Where e comes from

Imagine a bank that pays interest equal to your whole balance once a year: a dollar becomes \$2.00. Split that interest into two half-yearly payments and you get \$2.25, because the second payment earns interest on the first. Pay it more and more often:

A dollar with interest equal to the balance, split into equal payments.

PAID HOW MANY TIMES A YEAR	\\$1 GROWS TO
1	2.00000
2	2.25000
4	2.44141
12	2.61304
365	2.71457
1,000,000	2.71828

Each step adds a little less, and the total never passes one number, $e = 2.71828\dots$ In symbols,

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

The same limit works for any rate. At a rate r compounded n times a year for T years, a dollar grows to $(1 + r/n)^{nT}$, and as n grows this tends to e^{rT} . That is **continuous compounding**, and it is how every formula in this series grows money.

Our rate

For the running rate of 3.6%:

A dollar at 3.6% for one year.

INTEREST PAID	\\$1 GROWS TO
once a year	1.03600
quarterly	1.03649
monthly	1.03660
daily	1.03665
continuously	1.03666

Paid once a year, a dollar becomes 1.0360; continuously compounded, 1.0367. The difference is small, 0.07 cents per \$100, but continuous compounding makes the maths much tidier. At that rate, \$100 today grows to \$103.67 in a year: the forward price of Chapter 4.

Discounting

Run it backwards. A dollar paid in T years is worth e^{-rT} today, the **discount factor**. At our rate, a dollar in a year is worth 0.9646, about 96.46 cents. Every option price in this book ends with this step: work out what you will receive, then bring it back to today. QuantLib's flat, continuously compounded curve gives the same factor, 0.964640.

Logarithms and returns

The natural logarithm undoes e : $\ln(e^x) = x$. It turns growth *factors*, which multiply, into *rates* that add.

$$\ln(a) + \ln(b) = \ln(ab)$$

Say a share rises 10% and then falls 10%. The percentages seem to cancel, but the share ends 1% down. In logs the two moves are 0.0953 and -0.1054, and they add up exactly to the log of the total change, -0.0101. That is why volatility, in Chapter 9, is measured on **log returns**.

Where they turn up next

- e^{rT} gives the forward price (Chapter 4).
- e^{-rT} discounts every payoff to today.

- $\ln(S/K)$ sits inside the Black-Scholes formula (Chapter 10), and log returns measure volatility (Chapter 9).

Bip's rule. e^{rT} grows it; e^{-rT} brings it home.

Try this. What is \$103.67, paid in a year, worth today at 3.6%?

Companion notebook: `s3e02_exp_and_log.ipynb` . Every number in this chapter is computed by it.

Why the Price Is Not the Expected Value

The most important idea in option pricing. People who disagree about the odds still have to agree on one price, or someone gets free money.

Last chapter's question

\$103.67 paid in a year, discounted at our rate, is worth exactly \$100 today.

A toy world

Shrink the world to two outcomes. The share is \$100 today, and in a year it will be either \$120 or \$80. Our call, struck at \$105, then pays \$15 or nothing.

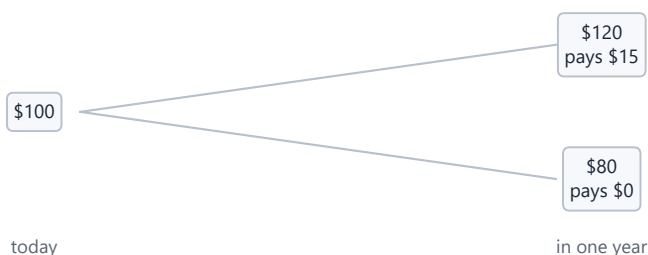


Figure 3.1. The toy world: two outcomes in a year, and what the call pays in each.

The toy is a simplification: Chapter 7 replaces it with a tree built from volatility. But the idea we find here carries over unchanged.

Two opinions, two prices

What is the call worth? The natural guess is to weigh each payoff by its chance and discount:

$$\text{price} \stackrel{?}{=} e^{-rT} [p C_u + (1 - p) C_d]$$

where p is the chance of the up move and C_u, C_d are the payoffs in the two states. The trouble is that people disagree about p . A pessimist who says 30% gets \$4.34; an optimist who says 80% gets \$11.58. Same option, very different prices. Neither is right.

Arbitrage

An **arbitrage** is a trade that needs none of your own money, carries no risk, and still leaves a profit. In a working market, anyone who spots one uses it until prices move and it disappears. So a price that allows arbitrage can't last.

Suppose someone pays the optimist's price for the call. Sell it to them, then copy the option yourself: buy 0.375 of a share for \$37.50 and borrow \$28.94. (Chapter 5 shows where 0.375 comes from.)

Selling the call at the optimist's price and copying it. Cash in (+) and out (–).

	CALL	SHARES	LOAN	TOTAL
today	+\$11.58	–\$37.50	+\$28.94	+\$3.01
share at \$120	–\$15.00	+\$45.00	–\$30.00	\$0.00
share at \$80	\$0.00	+\$30.00	–\$30.00	\$0.00

You pocket \$3.01 today. In a year the shares repay the loan of \$30.00 and cover whatever you owe on the call, in both outcomes: the totals at expiry are zero. The profit was free money.

It works the other way too. If someone sells the call at the pessimist's price, buy it and run the copy in reverse: sell the shares short and lend \$28.94. That leaves \$4.22 today and, again, nothing owed at expiry.

Buying the call at the pessimist's price and copying it in reverse.

	CALL	SHARES	LOAN	TOTAL
today	–\$4.34	+\$37.50	–\$28.94	+\$4.22
share at \$120	+\$15.00	–\$45.00	+\$30.00	\$0.00
share at \$80	\$0.00	–\$30.00	+\$30.00	\$0.00

The only price

Only one price gives nobody free money: the cost of the copy, \$8.56. At any other price one side of the trade above is a free lunch. The chart shows the "expected value" price for every opinion from 0% to 100%, against that single line.

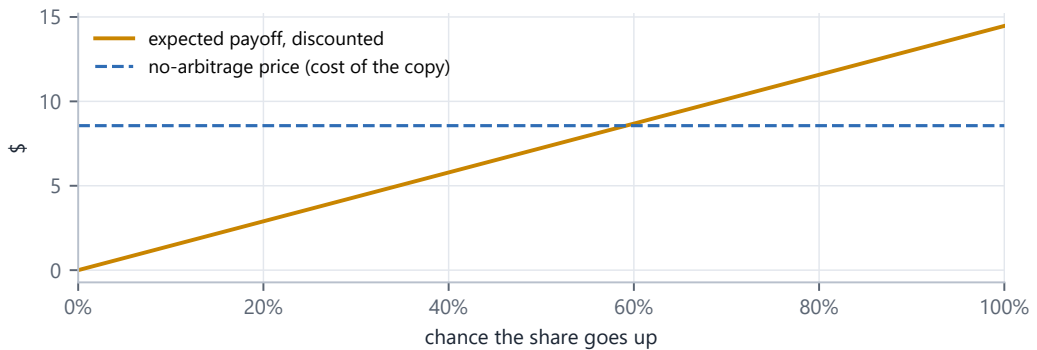


Figure 3.2. The expected-value price for every opinion about the up move, and the one price that rules out arbitrage.

The line is flat: opinions never enter it. Even if everyone were certain the share would rise, the fair price would stay \$8.56. An option's price is not a forecast of what it will pay. It is the cost of copying it.

Bip's rule. Price by copying, not by guessing.

Try this. If everyone became certain the share would rise, would the fair price change?

Companion notebook: `s3e03_price_not_expected_value.ipynb`. Every number in this chapter is computed by it.

Forwards and Put-Call Parity

The first price that needs no model at all, only the rule that there is no free money.

Last chapter's question

No. The price is the cost of copying the option, and nobody's view of the odds changes the copy.

The fair forward price

A **forward** is an agreement today to buy the share on a later date at a price fixed now. What price is fair for one year? Borrow \$100, buy the share today and hold it. In a year you own the share and owe \$103.67. So the forward price must be

$$F = S e^{rT} = 103.67$$

At any other forward price, someone locks in a profit: if the forward were higher, sell it and do the trade above; if lower, the reverse.

Two portfolios, one payoff

Buy the call and sell the put, both struck at \$105. Above the strike the call pays you $S_T - K$ and the put is worthless. Below it the call is worthless and you owe the put's buyer $K - S_T$. Either way you end up with exactly $S_T - K$: the payoff of a forward agreement to buy at the strike.

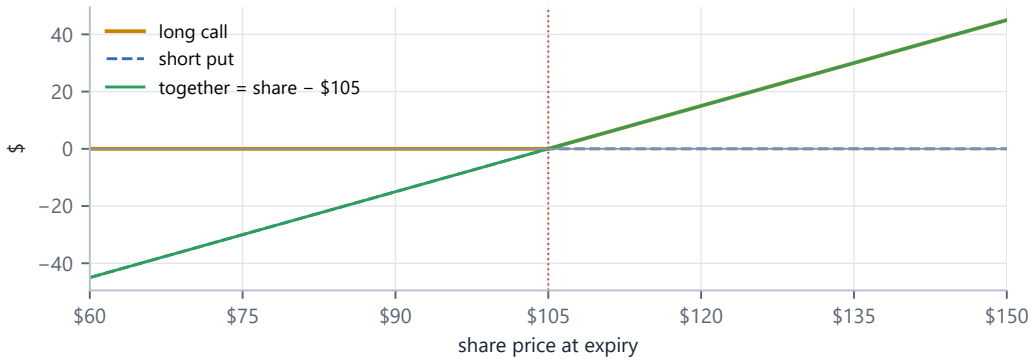


Figure 4.1. A long call plus a short put, both struck at \$105, pay the same as a forward to buy at \$105.

Put-call parity

Two positions with the same payoff at expiry must cost the same today. A forward to buy at K is worth the share minus the strike's present value, so

$$C - P = S - Ke^{-rT}$$

That is **put-call parity**. It has no volatility in it, no model, and no opinions. It holds for any strike and any volatility; the notebook checks several of each.

For our trade, the call is \$7.39 and the put \$8.68, so $C - P$ is $-\$1.29$. The share is \$100 and the strike's present value is \$101.29, so $S - Ke^{-rT}$ is $-\$1.29$. They match.

Why the put costs more

Parity also says the call and the put cost the same exactly when the strike equals the forward price:

$$C = P \iff K = Se^{rT}$$

Call and put by strike. The middle strike is the forward price.

STRIKE	CALL	PUT
\$100.00	\$9.72	\$6.18
\$103.67	\$7.97	\$7.97
\$105.00	\$7.39	\$8.68

Below the forward the call is dearer; at the forward both cost \$7.97; above it the put is dearer. Our strike, \$105, is above the forward, so the put costs more. It isn't fear. It is where the strike sits.

Catching a mispriced put

Parity also catches mistakes. Suppose someone quotes our put at \$8.18, too cheap. Sell the call, buy the put and the share, and borrow the strike's present value, \$101.29.

The parity arbitrage with the put quoted at \$8.18. Cash in (+) and out (–).

	CALL	PUT	SHARE	LOAN	TOTAL
today	+\$7.39	–\$8.18	–\$100.00	+\$101.29	+\$0.50
share at \$80	\$0.00	+\$25.00	+\$80.00	–\$105.00	\$0.00
share at \$105	\$0.00	\$0.00	+\$105.00	–\$105.00	\$0.00
share at \$130	–\$25.00	\$0.00	+\$130.00	–\$105.00	\$0.00

You keep \$0.50 today. At expiry share plus put minus call is worth exactly the strike, whatever happens, which repays the loan.

In the notebook

QuantLib's analytic engine gives the same call and put, a forward of 103.6656, and a parity residual of zero.

Bip's rule. Same payoff, same price.

Try this. If interest rates rose, would our call get dearer or cheaper compared with the put?

Companion notebook: `s3e04_put_call_parity.ipynb`. Every number in this chapter is computed by it.

The One-Step Tree

Build a portfolio of shares and borrowed cash that pays exactly what the option pays. Its cost is the option's price, and its number of shares is the option's delta.

Last chapter's question

Dearer. Parity says $C - P = S - Ke^{-rT}$, and a higher rate shrinks Ke^{-rT} . At 5% instead of our rate, $C - P$ moves from $-\$1.29$ to $+\$0.12$.

The recipe

Back to the toy world: \$100 today, \$120 or \$80 in a year, and a call paying \$15 or nothing. We want a portfolio that pays the same as the call in both outcomes.

Step one: how many shares? Between the two outcomes the share's price moves by $S_u - S_d$ and the call's payoff by $C_u - C_d$. Holding Δ shares makes the portfolio move with the option when

$$\Delta = \frac{C_u - C_d}{S_u - S_d}$$

Step two: how much to borrow? In the down state the shares are worth ΔS_d and the call pays C_d . Borrow today exactly what repays the difference in a year:

$$B = e^{-rT} (\Delta S_d - C_d)$$

The copy costs $\Delta S - B$.

Our numbers

Delta is \$15 over \$40: 0.375 of a share, costing \$37.50. In the down state those shares are worth \$30, so we borrow what repays that in a year, \$28.94. The copy costs \$8.56.

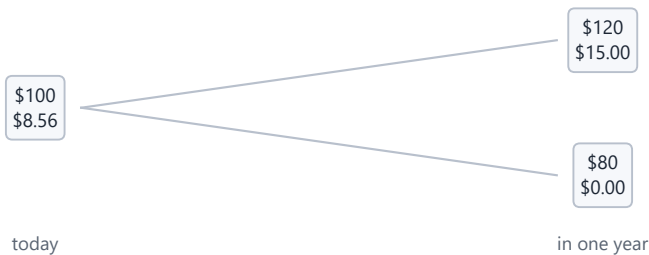


Figure 5.1. The one-step tree with the call's value in each state and, today, the cost of the copy.

Check both outcomes

The copy pays exactly what the call pays in both outcomes.

	SHARES WORTH	REPAY LOAN	COPY PAYS	CALL PAYS
share at \$120	\$45.00	−\$30.00	\$15.00	\$15.00
share at \$80	\$30.00	−\$30.00	\$0.00	\$0.00

Same payoff in every outcome, so the same price today: \$8.56, the number from Chapter 3.

Copying a put

The same recipe copies the put, which pays nothing up and \$25 down. Now delta is negative, -0.625: the copy *sells* 0.625 of a share short, worth \$62.50, and *lends* \$72.35. It costs \$9.85. And the call minus the put is −\$1.29, exactly the put-call parity of Chapter 4.

Delta

The number of shares that copies an option is its **delta**. It is also how many shares a dealer who sold the option holds to hedge it: short the call, long delta shares, and the two move together. On this tree delta falls as the strike rises:

Delta and price across strikes on the same one-step tree.

STRIKE	DELTA (SHARES)	PRICE
\$90	0.750	\$17.12
\$100	0.500	\$11.41
\$105	0.375	\$8.56
\$110	0.250	\$5.71
\$120	0.000	\$0.00

At a strike of \$120 or above the call never pays, and delta is zero. We meet delta again in Chapter 10, and all through Part 2.

In the notebook

The one-step copy in options.py

```
def one_step(s, up, down, k, t, r, call=True):
    """One-step tree: replicate the option with delta shares and cash. Returns the pieces and
    the price."""
    pay = call_payoff if call else put_payoff
    v_up, v_down = float(pay(up, k)), float(pay(down, k))
    delta = (v_up - v_down) / (up - down)
    cash = (v_down - delta * down) * math.exp(-r * t) # negative: borrowed today
    q = (s * math.exp(r * t) - down) / (up - down) # risk-neutral probability of the up
    move
    return {"v_up": v_up, "v_down": v_down, "delta": delta, "cash": cash, "price": delta * s
    + cash, "q": q}
```

Bip's rule. Delta is how many shares copy the option.

Try this. On the same tree, what is delta for a call struck at \$80?

Companion notebook: `s3e05_one_step_tree.ipynb`. Every number in this chapter is computed by it.

Risk-Neutral Probability

One special set of odds turns "expected payoff, discounted" into the right price. It is a calculating device, not anyone's forecast.

Last chapter's question

Delta is 1. A call struck at \$80 pays \$40 if the share goes up and nothing if it goes down: exactly the share's own swing. The copy is one whole share, less a loan of the present value of \$80, and the option moves one for one with the share.

The shortcut

Chapter 3 showed that weighting payoffs by people's odds gives the wrong price. But there is exactly one set of odds that gives the right one. Write the two outcomes as growth factors, $u = S_u/S$ and $d = S_d/S$, here 1.20 and 0.80. Then

$$q = \frac{e^{rT} - d}{u - d}$$

and the price is the q -weighted payoff, discounted:

$$C = e^{-rT} [q C_u + (1 - q) C_d]$$

For our tree, q is 59.16%, so $1 - q$ is 40.84%.

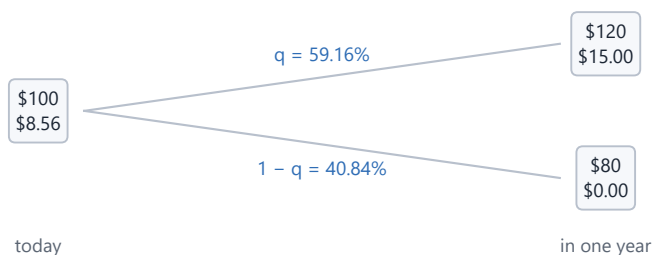


Figure 6.1. The one-step tree with the risk-neutral probabilities on its branches.

The average payoff under q is \$8.87; discounted, \$8.56. That is exactly the cost of the copy in Chapter 5, \$8.56. It has to be: substitute Δ and B from Chapter 5 into $\Delta S - B$ and rearrange, and you get the formula above. The shortcut is the copy written a different way.

Any payoff

The same q prices every payoff on this tree. The put pays \$25 only in the down state, so its price is that payoff times $1 - q$, discounted: \$9.85. That is also its copy price, \$9.85, and its put-call parity value, \$9.85.

Why "risk-neutral"

Under q , the share's average price in a year is \$103.67: exactly the forward price, \$103.67. The share grows at the risk-free rate, as if nobody needed to be paid for taking risk. That is all "risk-neutral" means.

Fake odds, real price

The real-world odds, the ones people argue about, don't enter the price:

Different views of the share, one option price.

	THEIR CHANCE OF UP	THEIR EXPECTED SHARE PRICE	OPTION PRICE
pessimist	30%	\$92.00	\$8.56
optimist	80%	\$112.00	\$8.56

q is not a forecast. It is forced on us by the up move, the down move and the interest rate. And "risk-neutral" doesn't mean investors ignore risk. As the Royal Swedish Academy of Sciences put it when awarding the 1997 prize for this work, no risk premium is needed to value an option because it is already included in the share price. Every model from here to the end of the series prices this way: average under risk-neutral odds, then discount.

When rates change

Higher rates raise the forward, so q must rise to make the share grow faster on average, and the call gets dearer:

The risk-neutral probability and the call price for three interest rates, same tree.

INTEREST RATE (%)	Q (%)	CALL
0.0	50.00	\$7.50
3.6	59.16	\$8.56
5.0	62.82	\$8.96

Bip's rule. Risk-neutral odds are a pricing tool, not a forecast.

Try this. If interest rates rose, would q go up or down?

Sources

Companion notebook: `s3e06_risk_neutral_probability.ipynb`. Every number in this chapter is computed by it.

1. The Royal Swedish Academy of Sciences (nobelprize.org), *Press release: The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 1997*. <https://www.nobelprize.org/prizes/economic-sciences/1997/press-release/>

Many Steps, and American Options

Real trees size their steps from volatility, and we value options by working backward from expiry. With a right to exercise early, check at every node whether exercising beats holding.

Last chapter's question

Up. $q = (e^{rT} - d)/(u - d)$ grows with r : at 5%, q rises from 59.16% to 62.82%.

Steps sized by volatility

Real trees have many small steps. The standard recipe, from Cox, Ross and Rubinstein (1979), sizes each step of length Δt from the volatility σ :

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = \frac{1}{u}, \quad q = \frac{e^{r\Delta t} - d}{u - d}$$

Chapter 9 explains the square root. With our 20% volatility and two six-month steps, u is 1.1519, d is 0.8681 and q is 0.5287.

Working backward

Build the tree forward, then value it backward. At expiry the call pays its payoff. One step earlier, each node is worth the q -weighted average of the two nodes after it, discounted over one step. Repeat back to today.

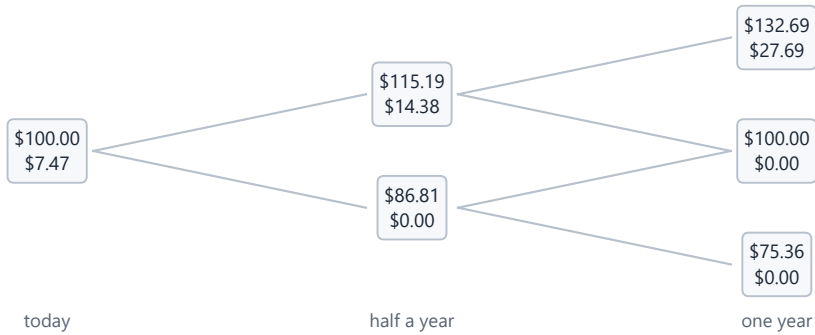


Figure 7.1. A two-step tree for the one-year \$105 call: share price above, call value below.

The call pays only at the top, \$27.69. The node above it after six months is worth \$14.38, and today the call is worth \$7.47. Add more steps and the price settles down: \$7.55 with ten steps, \$7.40 with a hundred, \$7.39 with five hundred.

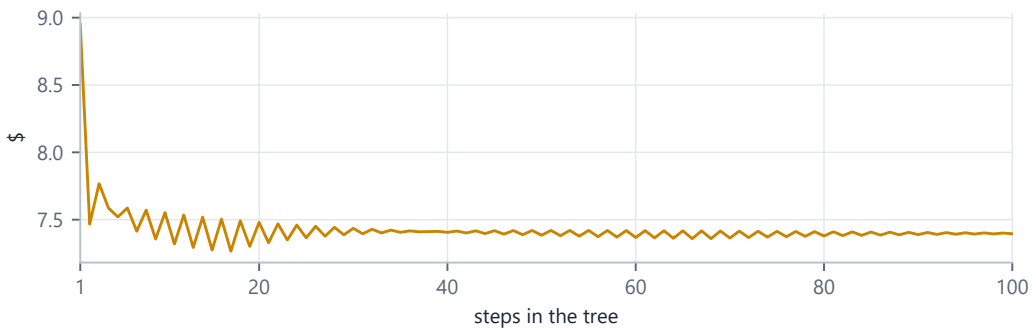


Figure 7.2. The tree price of the call as the number of steps grows.

Chapter 10 shows what it is settling on.

European and American

So far our options could be used only at expiry. Those are **European** options. **American** options can be exercised on any day up to expiry. ASX stock options are usually American, and index options European.

For an American option, the backward step asks one more question at every node: is exercising now worth more than holding on? The node's value is the larger of the two.

An American put

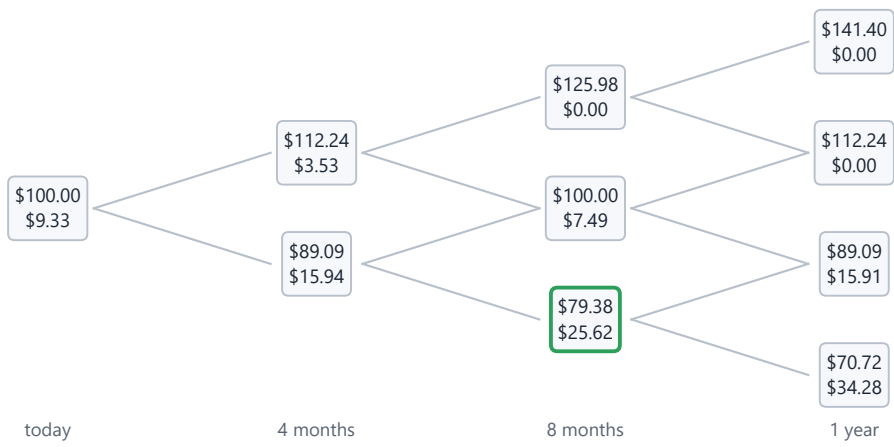


Figure 7.3. An American put on a three-step tree. The green node is where exercising early beats holding.

At the lowest node after eight months, the share is at \$79.38. Exercising pays \$25.62 at once, more than holding the option is worth: the cash starts earning interest, and with the share already low there is little left to gain by waiting. On this tree the American put is worth \$9.33, against \$9.05 for the European one.

The value of the right

With five hundred steps the American put is worth \$9.24 and the European \$8.68, so the right to exercise early is worth \$0.57. The tree also says when to use it: the **exercise boundary**, the share price below which exercising is best.

Exercise boundary of the American \$105 put, 500-step tree.

MONTHS TO EXPIRY	EXERCISE IF THE SHARE IS BELOW
9	\$82.88
6	\$85.13
3	\$89.02

Calls are different

On a share that pays no dividends, exercising a call early never pays: you would hand over the strike sooner and give up the protection the call still gives. The American call is worth exactly the European one, \$7.39.

In the notebook

QuantLib's Cox-Ross-Rubinstein engine uses the same u and d but a slightly different up-probability, $\frac{1}{2} + \frac{1}{2}(r - \sigma^2/2)\sqrt{\Delta t}/\sigma$, a linearised version of ours. Both converge to the same price. With QuantLib's probability, the hand tree matches QuantLib: \$9.24 and \$9.24 for the American put.

The n-step tree in options.py, European or American

```
def binomial_tree(s, k, t, r, vol, steps, call=True, american=False, quantlib_p=False):
    """Cox-Ross-Rubinstein tree: u = exp(vol sqrt(dt)), d = 1/u, worked backward from expiry.
    quantlib_p=True uses QuantLib's up-probability 1/2 + 1/2 (r - vol^2/2) sqrt(dt) / vol instead of the textbook
    (e^{r dt} - d) / (u - d); both give the same price in the limit."""
    dt = t / steps
    u = math.exp(vol * math.sqrt(dt)); d = 1 / u
    p = 0.5 + 0.5 * (r - vol ** 2 / 2) * math.sqrt(dt) / vol if quantlib_p else (math.exp(r
* dt) - d) / (u - d)
    pay = call_payoff if call else put_payoff
    prices = s * d ** np.arange(steps, -1, -1) * u ** np.arange(steps + 1)
    values = pay(prices, k)
    for _ in range(steps):
        prices = prices[1:] / u
        values = math.exp(-r * dt) * (p * values[1:] + (1 - p) * values[:-1])
        if american:
            values = np.maximum(values, pay(prices, k))
    return float(values[0])
```

Bip's rule. Work backward: at each node, hold or exercise, whichever is worth more.

Try this. Why does the put's exercise boundary rise as expiry gets closer?

Sources

Companion notebook: `s3e07_many_steps_american.ipynb`. Every number in this chapter is computed by it.

1. ASX, *Options course, Module 2: What are options*. <https://www.asx.com.au/investors/investment-tools-and-resources/online-courses/options-course-quiz/what-are-options>
2. Elsevier, *Option pricing: A simplified approach (Cox, Ross and Rubinstein)*, *Journal of Financial Economics* 7(3), 229-263. [https://doi.org/10.1016/0304-405X\(79\)90015-1](https://doi.org/10.1016/0304-405X(79)90015-1)
3. QuantLib, *QuantLib 1.43 (BinomialCRRVanillaEngine, CoxRossRubinstein tree)*. <https://www.quantlib.org/>

Maths Pit Stop: the Bell Curve

Many small random steps add up to a bell curve. Its centre is the mean, its width the standard deviation, and $N(x)$ measures the area to the left of x .

Last chapter's question

With less time left, there is less chance the share climbs back above the strike, so waiting is worth less, and exercising wins at higher share prices. That is why the boundary rises as expiry nears.

Coin flips make a bell

Flip a fair coin n times and count the heads. The exact chance of k heads is

$$P(k) = \binom{n}{k} \frac{1}{2^n}$$

With 10 flips, 5 heads is the most likely count, but anything from none to all of them can happen. With 100 and then 1,000 flips the same shape appears, and it matches a smooth curve better and better: the **normal distribution**, or bell curve. At 1,000 flips the largest gap between the exact chances and the curve is $6.3\text{e-}06$.

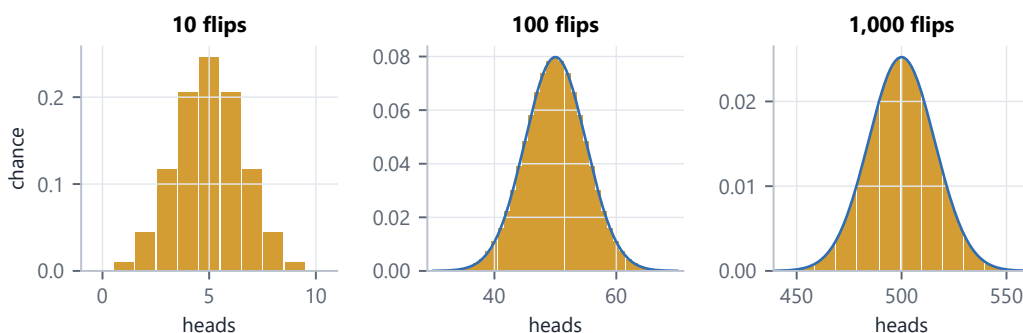


Figure 8.1. Exact chances of each number of heads for 10, 100 and 1,000 fair coin flips, with the normal curve.

Adding up many small random steps gives a bell curve, whatever the individual steps look like. That is why it turns up in option pricing: a share price is the result of many small moves.

Mean and standard deviation

Two numbers describe a bell curve. The **mean** is its centre; the **standard deviation** measures its spread. For coin flips the mean is $n/2$ and the standard deviation is $\sqrt{n}/2$.

Mean and standard deviation of the number of heads.

FLIPS	MEAN	STANDARD DEVIATION
10	5.0	1.581
100	50.0	5.000
1,000	500.0	15.811

Ten times as many flips spread the count only 3.16 times wider. Spread grows with the square root of the number of steps, a rule that returns in Chapter 9 as the square-root-of-time rule for volatility.

The rule of thumb

For any normal distribution, about 68.3% of outcomes lie within one standard deviation of the mean, 95.4% within two, and 99.7% within three. So an outcome three standard deviations out should be very rare. Part 3 shows that real markets produce more of them than the bell curve allows.

$N(x)$

$N(x)$ is the chance that a standard normal outcome (mean 0, standard deviation 1) is below x : the area under the bell to the left of x .

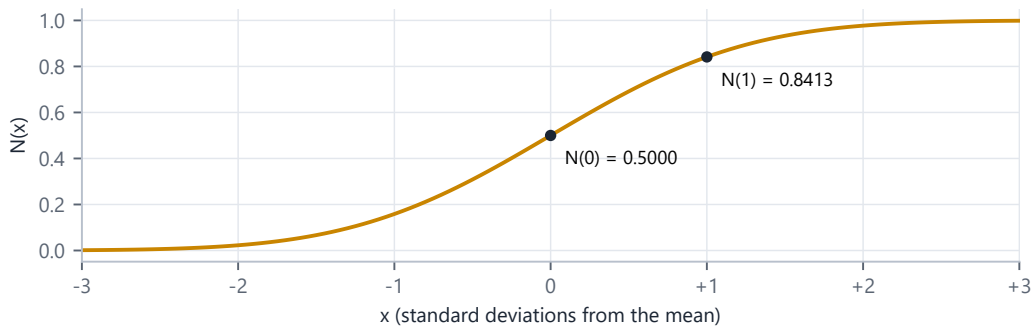


Figure 8.2. The cumulative normal distribution $N(x)$.

$N(0)$ is 0.50, exactly half, and $N(1)$ is 0.8413. By symmetry $N(-x) = 1 - N(x)$, so $N(-1)$ is 0.1587. It is the N in the Black-Scholes formula, $N(d_1)$ and $N(d_2)$, and Chapter 10 shows exactly what each measures. The notebook computes it from the error function, and it matches QuantLib's `CumulativeNormalDistribution` to 2.2e-16.

Bip's rule. Spread grows with the square root of the steps.

Try this. Flip 400 coins. What is the standard deviation of the number of heads?

Companion notebook: `s3e08_bell_curve.ipynb`. Every number in this chapter is computed by it.

What Volatility Is

A tree with many steps becomes a lognormal distribution. Volatility is the yearly standard deviation of log returns, and it scales with the square root of time. Then, real data.

Last chapter's question

The standard deviation of the number of heads in 400 flips is $\sqrt{400}/2 = 10$.

A tree turns into a bell curve

Take the tree of Chapter 7 with 250 steps, about one per trading day, and look at where the share can end up in a year and how likely each end price is under the risk-neutral odds. Each end price, with its chance spread over the dollars around it, sits on a smooth curve: the **lognormal** distribution, the shape you get when the *logarithm* of the price is normally distributed.

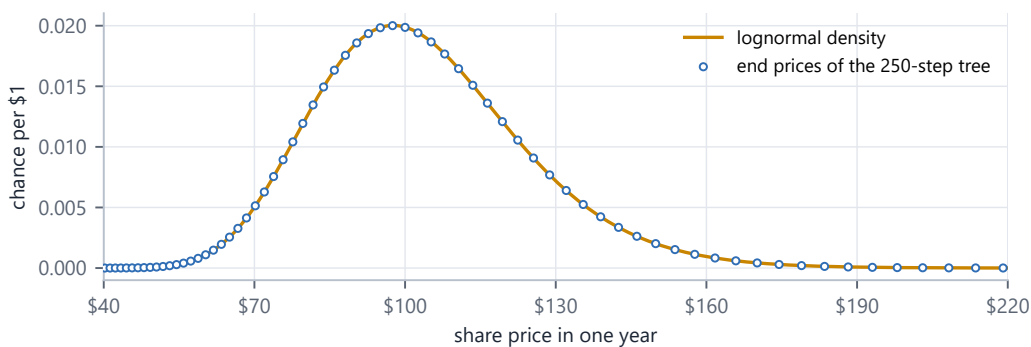


Figure 9.1. End prices of a 250-step tree (dots) against the lognormal density. Illustrative share, 20% volatility.

The curve is lopsided, with a long tail to the right and nothing below zero. Its average is exactly the forward price, \$103.67: the tree's odds are risk-neutral.

Why the square root?

Each day's move is a separate, independent random step, like a coin flip. For independent steps it is the *variances* that add up, not the standard deviations:

$$\text{Var}(X_1 + \cdots + X_n) = n \text{Var}(X)$$

so the standard deviation over n days is $\sqrt{n}$ times one day's. That is Chapter 8's coin-flip rule, and it is why the tree's steps in Chapter 7 are sized with $\sqrt{\Delta t}$.

Volatility, defined

Volatility, σ , is the standard deviation of the share's log returns over a year. With about 252 trading days in a year, a daily move is typically

$$\sigma_{\text{day}} = \frac{\sigma}{\sqrt{252}}$$

Our 20% share moves about 1.26% on a typical day.

A lopsided range

One standard deviation either side, in log terms, takes the share to $Se^{\pm\sigma\sqrt{T}}$:

One standard deviation either side of \$100, at 20% volatility.

	LOW	HIGH	DOWN BY	UP BY
one year	\$81.87	\$122.14	\$18.13	\$22.14
three months	\$90.48	\$110.52	\$9.52	\$10.52

Over a year the share goes \$22.14 up but only \$18.13 down: a price can double but can't fall below zero. Over three months $\sqrt{T} = 1/2$, so the range halves in log terms.

Real data: the Australian dollar

The Federal Reserve Board publishes the exchange rate of the Australian dollar against the US dollar each day: the noon buying rate in New York. Over the 252 trading days to 18 September 2026, the standard deviation of its daily log changes was 0.48%. Times $\sqrt{252}$, that is a **realised volatility** of 7.5%, much calmer than our share.

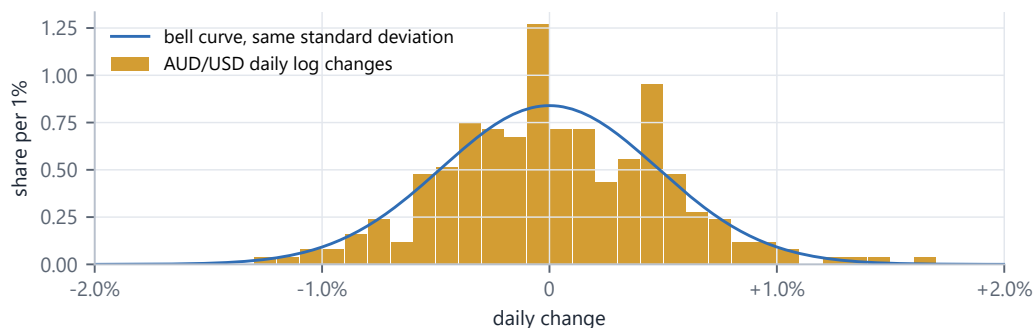


Figure 9.2. AUD/USD daily log changes over the year to 18 Sep 2026, against a bell curve with the same standard deviation. Source: Board of Governors of the Federal Reserve System, H.10.

The bell curve fits the middle well. The tails are a different story, and Part 3 is about them.

Volatility moves

Measured over rolling three-month windows since 2018, AUD/USD volatility has been as low as 5.8% and as high as 20.1%, in the three months to June 2020. Over the whole period it was 10.0%.



Figure 9.3. AUD/USD realised volatility over rolling three-month (63-day) windows, annualised. Source: Board of Governors of the Federal Reserve System, H.10.

The Black-Scholes formula assumes one constant volatility. The data say it changes a lot. That assumption is what Part 3 takes apart.

Bip's rule. Volatility scales with the square root of time.

Try this. A share has 16% volatility. Roughly how big is a typical daily move?

Sources

Companion notebook: `s3e09_what_is_volatility.ipynb`. Every number in this chapter is computed by it.

1. Board of Governors of the Federal Reserve System, *Website disclaimer (copyright and public domain)*. <https://www.federalreserve.gov/disclaimer.htm>
2. Board of Governors of the Federal Reserve System, *About the H.10 release*. <https://www.federalreserve.gov/releases/h10/about.htm>
3. Board of Governors of the Federal Reserve System, *H.10 Foreign Exchange Rates, daily data download (Australian dollar, series RXI\$US_N.B.AL)*. <https://www.federalreserve.gov/releases/h10/>

The Black-Scholes Formula, Decoded

Why our call is worth exactly \$7.39. What you receive minus what you pay, each weighted by its risk-neutral chance.

Last chapter's question

A typical daily move is the volatility divided by the square root of the trading days in a year: 16% over $\sqrt{252}$ is about 1.01%.

The formula

For a European call on a share that pays no dividends:

$$C = S N(d_1) - K e^{-rT} N(d_2)$$

with

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2) T}{\sigma \sqrt{T}}, \quad d_2 = d_1 - \sigma \sqrt{T}$$

N is the cumulative normal distribution of Chapter 8. The formula has five inputs:

The inputs, and d_1 and d_2 for our trade.

	OUR TRADE
Share price	\$100
Strike	\$105
Time to expiry	1 year
Interest rate	3.60%
Volatility	20%
d_1	0.0360
d_2	-0.1640

N(d2): what you pay

$N(d_2)$, here 0.4349, is the chance under the risk-neutral odds of Chapter 6 that the call ends in the money, so that you pay the strike. A million simulated risk-neutral outcomes agree: 43.5% of them end above the strike. The cash side is the strike's present value times that chance:

$$Ke^{-rT}N(d_2) = 101.29 \times 0.4349 = 44.05$$

N(d1): what you get

The share side is $S N(d_1) = 51.44$. And $N(d_1)$, 0.5144, is the call's **delta**: the number of shares that copies the option, as in Chapter 5, now for a tree with infinitely many steps. Nudge the share price by a cent either way and the call moves by 0.5144 per dollar. $N(d_1)$ is *not* the probability of exercise; that is $N(d_2)$.

The price

$$51.44 - 44.05 = 7.39$$

That is the answer to Chapter 1's question, and QuantLib's analytic engine gives the same \$7.39, with delta 0.5144.

The Black-Scholes formula in options.py

```
def black_scholes(s, k, t, r, vol, call=True):
    """Black-Scholes price of a European option on a share with no dividends (r continuously
    compounded)."""
    d1 = (math.log(s / k) + (r + vol ** 2 / 2) * t) / (vol * math.sqrt(t))
    d2 = d1 - vol * math.sqrt(t)
    if call:
        return s * norm_cdf(d1) - k * math.exp(-r * t) * norm_cdf(d2)
    return k * math.exp(-r * t) * norm_cdf(-d2) - s * norm_cdf(-d1)
```

The tree agrees

The binomial tree of Chapter 7 converges to the formula. Odd and even step counts land on opposite sides of it, and the wobble shrinks as the steps shrink.

Tree prices of the call against the Black-Scholes price, \$7.39.

STEPS	TREE PRICE	TREE - FORMULA
10	\$7.55	0.1628
11	\$7.32	-0.0705
100	\$7.40	0.0057
101	\$7.40	0.0101
1,000	\$7.39	-0.0002

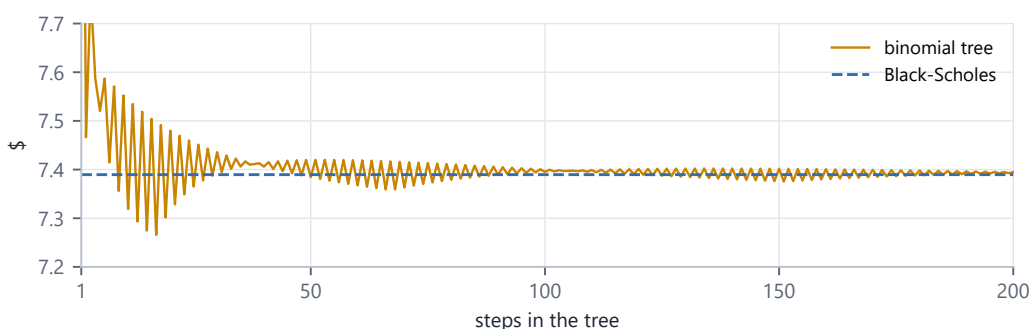


Figure 10.1. The binomial tree price as the number of steps grows, against the Black-Scholes price.

Black-Scholes is simply the tree taken to the limit.

What isn't in the formula

The share's expected return is not an input. It dropped out in Chapter 3 and again in Chapter 6, because the risk premium is already inside the share price. Everything else you can look up, except volatility:

The call price for three volatilities, all else as our trade.

VOLATILITY (%)	CALL
10	\$3.40
20	\$7.39
30	\$11.37

Where that one number comes from is where Parts 2 and 3 begin.

A little history

Fischer Black and Myron Scholes published the formula in 1973, and Robert Merton found another way to derive it and generalised it. Cox, Ross and Rubinstein's binomial tree followed in 1979. In 1997 Merton and Scholes received the prize in economic sciences; Black had died in 1995.

Bip's rule. Black-Scholes is the tree, taken to the limit.

Try this. The Part 1 checkpoint in the appendix.

Sources

Companion notebook: `s3e10_black_scholes_decoded.ipynb`. Every number in this chapter is computed by it.

1. Elsevier, *Option pricing: A simplified approach (Cox, Ross and Rubinstein)*, *Journal of Financial Economics* 7(3), 229-263. [https://doi.org/10.1016/0304-405X\(79\)90015-1](https://doi.org/10.1016/0304-405X(79)90015-1)
2. The Royal Swedish Academy of Sciences (nobelprize.org), *Press release: The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 1997 (14 October 1997)*. <https://www.nobelprize.org/prizes/economic-sciences/1997/press-release/>

Checkpoint and glossary

Part 1 checkpoint

Try these before Part 2. The answers follow.

1. Price the one-year \$105 call on a one-step tree where the share goes from \$100 to \$130 or \$75, first by copying it with shares and a loan, then with risk-neutral odds. The rate is 3.60%.
2. A \$100-strike call costs \$9.72 and the put \$6.18, both one year. Is put-call parity satisfied at 3.60%?
3. For a share expected to grow faster than the risk-free rate, which is bigger: the real-world or the risk-neutral chance that the call is exercised?

Answers.

1. The call pays \$25 or nothing, so delta is $25/55 = 0.4545$ shares. The copy costs \$12.57. The risk-neutral probability is 0.5212, and $e^{-rT} \times q \times 25$ gives the same \$12.57.
2. Yes. Call minus put is \$3.54, and the share minus the strike's present value is \$3.54.
3. The real-world chance. A share that grows faster than the risk-free rate on average ends above the strike more often than it does under the risk-neutral odds, where it grows at exactly the risk-free rate.

Glossary

American option. Can be exercised on any day up to expiry.

Arbitrage. A trade that needs none of your own money, carries no risk, and leaves a profit.

Break-even. The share price at expiry where an option's profit is zero: strike plus premium for a call, strike minus premium for a put.

Call. The right, but not the obligation, to buy at the strike up to expiry.

Delta. The number of shares that copies, and so hedges, an option. For a call in Black-Scholes, $N(d_1)$.

Discount factor. Today's value of a dollar paid later: e^{-rT} with continuous compounding.

European option. Can be exercised only at expiry.

Exercise boundary. For an American put, the share price below which exercising early is best.

Forward price. The fair price, agreed today, to buy the share on a later date: Se^{rT} for a share with no dividends.

Log return. The natural logarithm of the ratio of two prices; log returns over consecutive periods add up.

Lognormal. The distribution of a price whose logarithm is normally distributed.

Payoff. What an option is worth at expiry. **Profit** is payoff minus premium.

Premium. The price the buyer pays for an option.

Put. The right, but not the obligation, to sell at the strike up to expiry.

Put-call parity. $C - P = S - Ke^{-rT}$ for European options with the same strike and expiry.

Risk-neutral probability. The odds under which the discounted expected payoff equals the price; under them the share grows at the risk-free rate. A pricing tool, not a forecast.

Standard deviation. A measure of spread: the square root of the variance.

Strike. The fixed price at which an option lets its holder buy (call) or sell (put).

Taker, writer. ASX's terms for an option's buyer and seller.

Volatility. The standard deviation of log returns over a year.

Further reading

These books cover the same ground from different angles; none of this book's examples come from them.

- John C. Hull, *Options, Futures, and Other Derivatives*: the standard text, from trees to Black-Scholes and beyond.
- Sheldon Natenberg, *Option Volatility and Pricing*: options from a trader's point of view.

- Steven E. Shreve, *Stochastic Calculus for Finance I: The Binomial Asset Pricing Model*: the rigorous version of Chapters 3 to 7.
- Mark Joshi, *The Concepts and Practice of Mathematical Finance*: why the pricing works.